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Introduction

The purpose of this project is to understand book prices, ratings and stock information and to demonstrate a complete beginner-level data science workflow. The project uses Python, BeautifulSoup, Pandas, Matplotlib and Scikit-learn.

Data is required to be collected directly from the Books to Scrape website, with at least 200 books collected. The required fields include the book title, price in GBP, star rating and stock availability. The project then requires data cleaning, visualisation, Linear Regression and professional reporting.

Data Acquisition

BeautifulSoup was used to read the HTML structure of the Books to Scrape website. Requests was used to download each catalogue page. The scraper follows the next-page link until the catalogue has been collected. This approach makes it possible to collect more than the required 200 records.

For each book, the scraper extracts the title, price, star rating, stock availability and stock quantity where it is displayed. The information is placed into a Pandas DataFrame and saved as a CSV file.

Data Cleaning and Preparation

The scraped data was checked for duplicate book titles, missing values and incorrect data types. Price and rating were converted to numeric values so that mathematical calculations and machine learning could be performed. Titles were stripped of unnecessary spaces. Missing stock quantities were represented as zero because the model requires a numeric value.

Rows with missing price or rating values were removed because price is the target variable and rating is one of the selected model features. The cleaned dataset was saved separately from the raw dataset.

Data Visualisation

A bar chart shows the number of books for each rating level. A histogram shows the distribution of book prices. A bar chart compares average price by rating. A scatter plot examines the relationship between rating and price. An availability chart also summarises stock status.

The visualisations are useful because they make the numerical results easier to understand. The rating-versus-price analysis is particularly important because the project investigates whether book rating may help explain book price.

Predictive Modelling

A Linear Regression model was created to estimate book price. The target variable is Price_GBP. The selected features are Rating, Stock_Quantity and Catalogue_Page. The dataset is divided into training and testing data. Eighty percent of the records are used to train the model and 20% are used to test it. A fixed random state is used so that the results can be reproduced.

References

Beautiful Soup (n.d.) Beautiful Soup Documentation. Available at: https://www.crummy.com/software/BeautifulSoup/bs4/doc/

Scikit-learn (n.d.) Linear Regression Documentation. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html